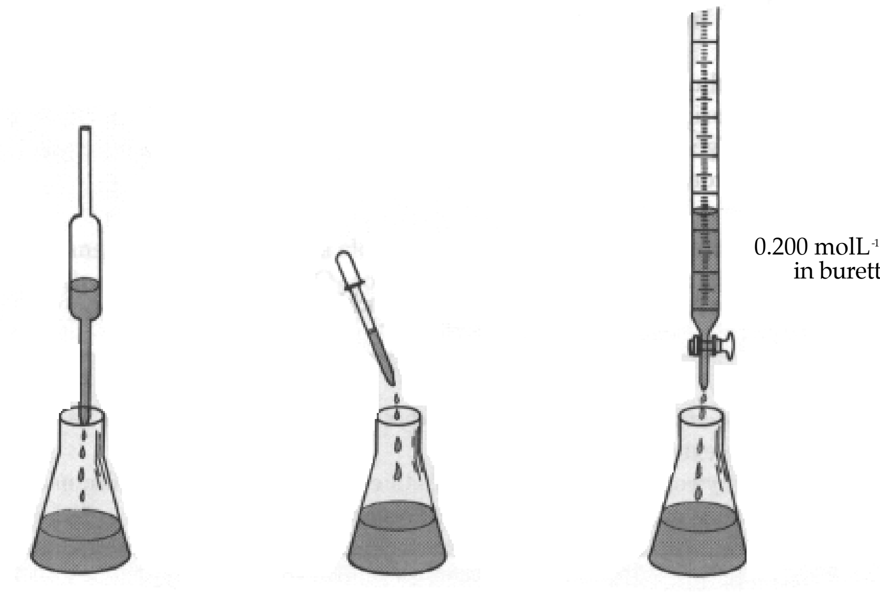
YEAR 10 BIOLOGICAL SCIENCE
BIOLOGICAL CHEMISTRY
TITRATIONS

AIM To determine the concentration of CO_2 in a carbonated drink.

METHOD

1. Set up a burette with some 0.200 $molL^{-1}$ NaOH solution.



Pipette out drink solution — Add phenolphthalein indicator — Titrate drink with NaOH

2. Record the initial level of NaOH accurately (to 2 decimal places).

3. Pipette 20.00 mL of the drink solution into a clean 250 mL conical flask and add 2-3 drops of phenolphthalein indicator.

4. Do a rough titration by running the NaOH into the flask while swirling it. Turn off the base when a colour change occurs. (This may be difficult to pick if the drink has its own colour.)

5. Record the final level of NaOH in the burette and determine the amount of base added.

6. Refill the burette if necessary with NaOH.

7. Prepare another 20.00 mL of drink solution and phenolphthalein in a clean conical flask.

8. Note the initial level of the NaOH in the burette. Run the NaOH into the flask to within 2.00 mL of the rough titration volume. Then add the acid drop-by-drop until a colour change occurs.

9. Record the final volume of the NaOH in the burette, and then the volume of NaOH added.

10. Repeat steps 7-9 for two other 20.00 mL samples of drink.

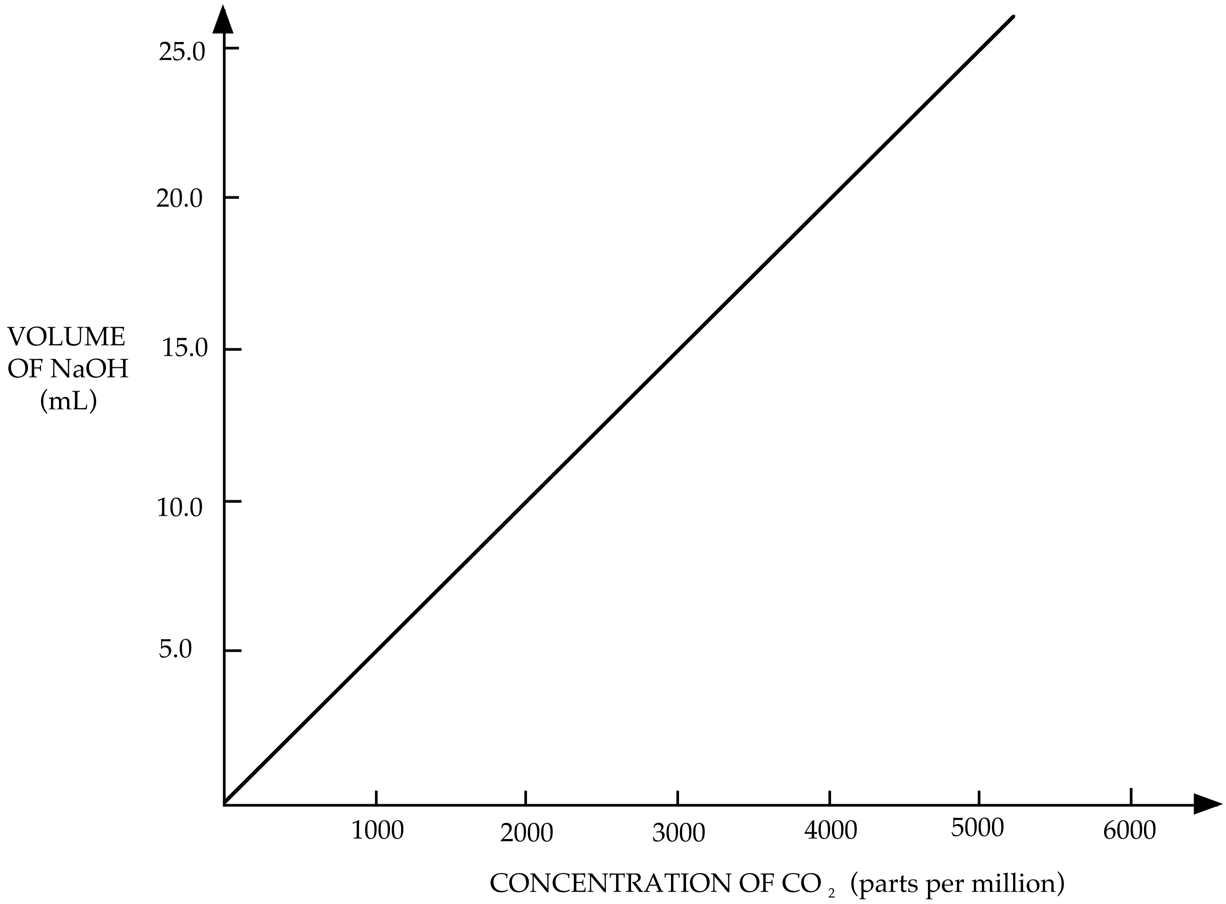
11. Average the accurate titration volumes to determine the amount of NaOH required to neutralise the drink solution.

12. Use the graph supplied to calculate the concentration of CO_2 in the drink.

RESULTS

VOLUME OF NaOH ADDED (mL)	ROUGH ESTIMATE	ACCURATE TITRATIONS		
		1	2	3
FINAL READING				
INITIAL READING				
TITRATION VOLUME				

AVERAGE TITRATION VOLUME ________________



CONCLUSION

CRITICAL ANALYSIS

QUESTIONS

1. (a) Would you expect a warm sample of the drink to have more or less CO_2 in it?

   (b) What does this tell you about the solubility of CO_2 in liquids?

2. Why does cold cool drink froth dramatically when poured into a warm glass from the cupboard?

3. Explain why it is not possible to use Coca-Cola in this experiment.

4. When carbonated drinks are canned, some air is removed and CO_2 is injected into the space above the liquid. Suggest a reason why the manufacturers do this.

5. (a) Why was NaOH chosen to titrate the drink solution rather than HCl?

   (b) What does this tell you about the pH of a carbonated solution?